Reproducibility statement

We provide our anonymised source code for reproducing all experiments at `https://anonymous.4open.science/r/tinylang-1061/`. We have specified the hyperparameters for training and evaluation as well as compute usage in §C and §4.1. We largely use open-source implementations of models from `zoology`[7] (Arora et al., 2024b) as well as the DeltaNet implementation from `fla`[8], with minor modifications for our mechanistic analysis with the open-source library `pyvene`[9] (Wu et al., 2024).

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References

Zeyuan Allen-Zhu and Alberto Alfarano. Physics of Language Models: Part 4.1, Architecture design and the magic of Canon layers. *SSRN*, 2025. URL https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5240330.

Simran Arora, Sabri Eyuboglu, Aman Timalsina, Isys Johnson, Michael Poli, James Zou, Atri Rudra, and Christopher Ré. Zoology: Measuring and improving recall in efficient language models. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net, 2024a. URL https://openreview.net/forum?id=LY3ukUANko.

Simran Arora, Sabri Eyuboglu, Michael Zhang, Aman Timalsina, Silas Alberti, James Zou, Atri Rudra, and Christopher Ré. Simple linear attention language models balance the recall-throughput tradeoff. In *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024*. OpenReview.net, 2024b. URL https://openreview.net/forum?id=e93ffDcpH3.

Jimmy Ba, Geoffrey E. Hinton, Volodymyr Mnih, Joel Z. Leibo, and Catalin Ionescu. Using fast weights to attend to the recent past. In Daniel D. Lee, Masashi Sugiyama, Ulrike von Luxburg, Isabelle Guyon, and Roman Garnett (eds.), *Advances in Neural Information Processing Systems 29: Annual Conference on Neural Information Processing Systems 2016, December 5-10, 2016, Barcelona, Spain*, pp. 4331–4339, 2016. URL https://proceedings.neurips.cc/paper/2016/hash/9f44e956e3a2b7b5598c625fcc802c36-Abstract.html.

Aviv Bick, Eric Xing, and Albert Gu. Understanding the skill gap in recurrent language models: The role of the gather-and-aggregate mechanism. *arXiv:2504.18574*, 2025. URL https://arxiv.org/abs/2504.18574.

Jannik Brinkmann, Abhay Sheshadri, Victor Levoso, Paul Swoboda, and Christian Bartelt. A mechanistic analysis of a transformer trained on a symbolic multi-step reasoning task. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar (eds.), *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 4082–4102, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-acl.242. URL https://aclanthology.org/2024.findings-acl.242/.

Ivo Danihelka, Greg Wayne, Benigno Uria, Nal Kalchbrenner, and Alex Graves. Associative long short-term memory. In Maria-Florina Balcan and Kilian Q. Weinberger (eds.), *Proceedings of*

[7] https://github.com/HazyResearch/zoology

[8] https://github.com/fla-org/flash-linear-attention

[9] https://github.com/stanfordnlp/pyvene

the 33nd International Conference on Machine Learning, ICML 2016, New York City, NY, USA, June 19-24, 2016, volume 48 of *JMLR Workshop and Conference Proceedings*, pp. 1986–1994. JMLR.org, 2016. URL http://proceedings.mlr.press/v48/danihelka16.html.

Tri Dao and Albert Gu. Transformers are SSMs: Generalized models and efficient algorithms through structured state space duality. *arXiv:2405.21060*, 2024. URL https://arxiv.org/abs/2405.21060.

Rares Dolga, Lucas Maystre, Marius Cobzarenco, and David Barber. Latte: Latent attention for linear time transformers. *arXiv:2402.17512*, 2024. URL https://arxiv.org/abs/2402.17512.

Ezra Edelman, Nikolaos Tsilivis, Benjamin L. Edelman, Eran Malach, and Surbhi Goel. The evolution of statistical induction heads: In-context learning markov chains. In Amir Globersons, Lester Mackey, Danielle Belgrave, Angela Fan, Ulrich Paquet, Jakub M. Tomczak, and Cheng Zhang (eds.), *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024*, Vancouver, BC, Canada, 2024. URL http://papers.nips.cc/paper_files/paper/2024/hash/75b0edb869e2cd509d64d0e8ff446bc1-Abstract-Conference.html.

Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021. URL https://transformer-circuits.pub/2021/framework/index.html.

Jiahai Feng and Jacob Steinhardt. How do language models bind entities in context? In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net, 2024. URL https://openreview.net/forum?id=zb3b6oKO77.

Daniel Y. Fu, Tri Dao, Khaled Kamal Saab, Armin W. Thomas, Atri Rudra, and Christopher Ré. Hungry hungry hippos: Towards language modeling with state space models. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023. URL https://openreview.net/forum?id=COZDy0WYGg.

Atticus Geiger, Hanson Lu, Thomas Icard, and Christopher Potts. Causal abstractions of neural networks. In M. Ranzato, A. Beygelzimer, Y. Dauphin, P.S. Liang, and J. Wortman Vaughan (eds.), *Advances in Neural Information Processing Systems*, volume 34, pp. 9574–9586. Curran Associates, Inc., 2021. URL https://proceedings.neurips.cc/paper_files/paper/2021/file/4f5c422f4d49a5a807eda27434231040-Paper.pdf.

Atticus Geiger, Duligur Ibeling, Amir Zur, Maheep Chaudhary, Sonakshi Chauhan, Jing Huang, Aryaman Arora, Zhengxuan Wu, Noah Goodman, Christopher Potts, and Thomas Icard. Causal abstraction: A theoretical foundation for mechanistic interpretability. *arXiv:2301.04709*, 2024. URL https://arxiv.org/abs/2301.04709.

Alex Graves, Greg Wayne, and Ivo Danihelka. Neural turing machines. *arXiv:1410.5401*, 2014. URL https://arxiv.org/abs/1410.5401.

Klaus Greff, Sjoerd van Steenkiste, and Jürgen Schmidhuber. On the binding problem in artificial neural networks. *arXiv:2012.05208*, 2020. URL https://arxiv.org/abs/2012.05208.

Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv:2312.00752*, 2024. URL https://arxiv.org/abs/2312.00752.

Samy Jelassi, David Brandfonbrener, Sham M. Kakade, and Eran Malach. Repeat after me: Transformers are better than state space models at copying. In *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024*. OpenReview.net, 2024. URL https://openreview.net/forum?id=duRRoGeoQT.

Najoung Kim and Sebastian Schuster. Entity tracking in language models. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki (eds.), *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 3835–3855, Toronto, Canada, July 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.acl-long.213. URL https://aclanthology.org/2023.acl-long.213/.

Belinda Z. Li, Zifan Carl Guo, and Jacob Andreas. (How) do language models track state? *arXiv:2503.02854*, 2025a. URL https://arxiv.org/abs/2503.02854.

Mingchen Li, Xuechen Zhang, Yixiao Huang, and Samet Oymak. On the power of convolution-augmented transformer. In Toby Walsh, Julie Shah, and Zico Kolter (eds.), *AAAI-25, Sponsored by the Association for the Advancement of Artificial Intelligence, February 25 - March 4, 2025, Philadelphia, PA, USA*, pp. 18393–18402. AAAI Press, 2025b. doi: 10.1609/AAAI.V39I17.34024. URL https://doi.org/10.1609/aaai.v39i17.34024.

Tom Lieberum, Matthew Rahtz, János Kramár, Neel Nanda, Geoffrey Irving, Rohin Shah, and Vladimir Mikulik. Does circuit analysis interpretability scale? evidence from multiple choice capabilities in chinchilla. *arXiv preprint arXiv:2307.09458*, 2023.

Bo Liu, Rui Wang, Lemeng Wu, Yihao Feng, Peter Stone, and Qiang Liu. Longhorn: State space models are amortized online learners. *arXiv:2407.14207*, 2024a. URL https://arxiv.org/abs/2407.14207.

Zicheng Liu, Siyuan Li, Li Wang, Zedong Wang, Yunfan Liu, and Stan Z. Li. Short-long convolutions help hardware-efficient linear attention to focus on long sequences. In *Forty-first International Conference on Machine Learning, ICML 2024*, Vienna, Austria, 2024b. OpenReview.net. URL https://openreview.net/forum?id=TRrXkVdhwi.

Shahar Lutati, Itamar Zimerman, and Lior Wolf. Focus your attention (with adaptive IIR filters). In Houda Bouamor, Juan Pino, and Kalika Bali (eds.), *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 12538–12549, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.772. URL https://aclanthology.org/2023.emnlp-main.772/.

Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. In *The Eleventh International Conference on Learning Representations, ICLR 2023*, Kigali, Rwanda, 2023. OpenReview.net. URL https://openreview.net/forum?id=9XFSbDPmdW.

Franz Nowak and Ryan Cotterell. A fast algorithm for computing prefix probabilities. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki (eds.), *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pp. 57–69, Toronto, Canada, July 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.acl-short.6. URL https://aclanthology.org/2023.acl-short.6/.

Destiny Okpekpe and Antonio Orvieto. Revisiting associative recall in modern recurrent models. In *First Workshop on Scalable Optimization for Efficient and Adaptive Foundation Models*, 2025. URL https://openreview.net/pdf?id=CcqAd5RPk5.

Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Scott Johnston, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. In-context learning and induction heads. *arXiv:2209.11895*, 2022. URL https://arxiv.org/abs/2209.11895.

Michael Poli, Stefano Massaroli, Eric Nguyen, Daniel Y. Fu, Tri Dao, Stephen Baccus, Yoshua Bengio, Stefano Ermon, and Christopher Ré. Hyena hierarchy: Towards larger convolutional language models. In Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett (eds.), *International Conference on Machine Learning, ICML 2023, 23-29 July 2023, Honolulu, Hawaii, USA*, volume 202 of *Proceedings of Machine Learning Research*, pp. 28043–28078. PMLR, 2023. URL https://proceedings.mlr.press/v202/poli23a.html.

Nikhil Prakash, Tamar Rott Shaham, Tal Haklay, Yonatan Belinkov, and David Bau. Fine-tuning enhances existing mechanisms: A case study on entity tracking. In *The Twelfth International Conference on Learning Representations, ICLR 2024*, Vienna, Austria, 2024. OpenReview.net. URL https://openreview.net/forum?id=8sKcAWOf2D.